

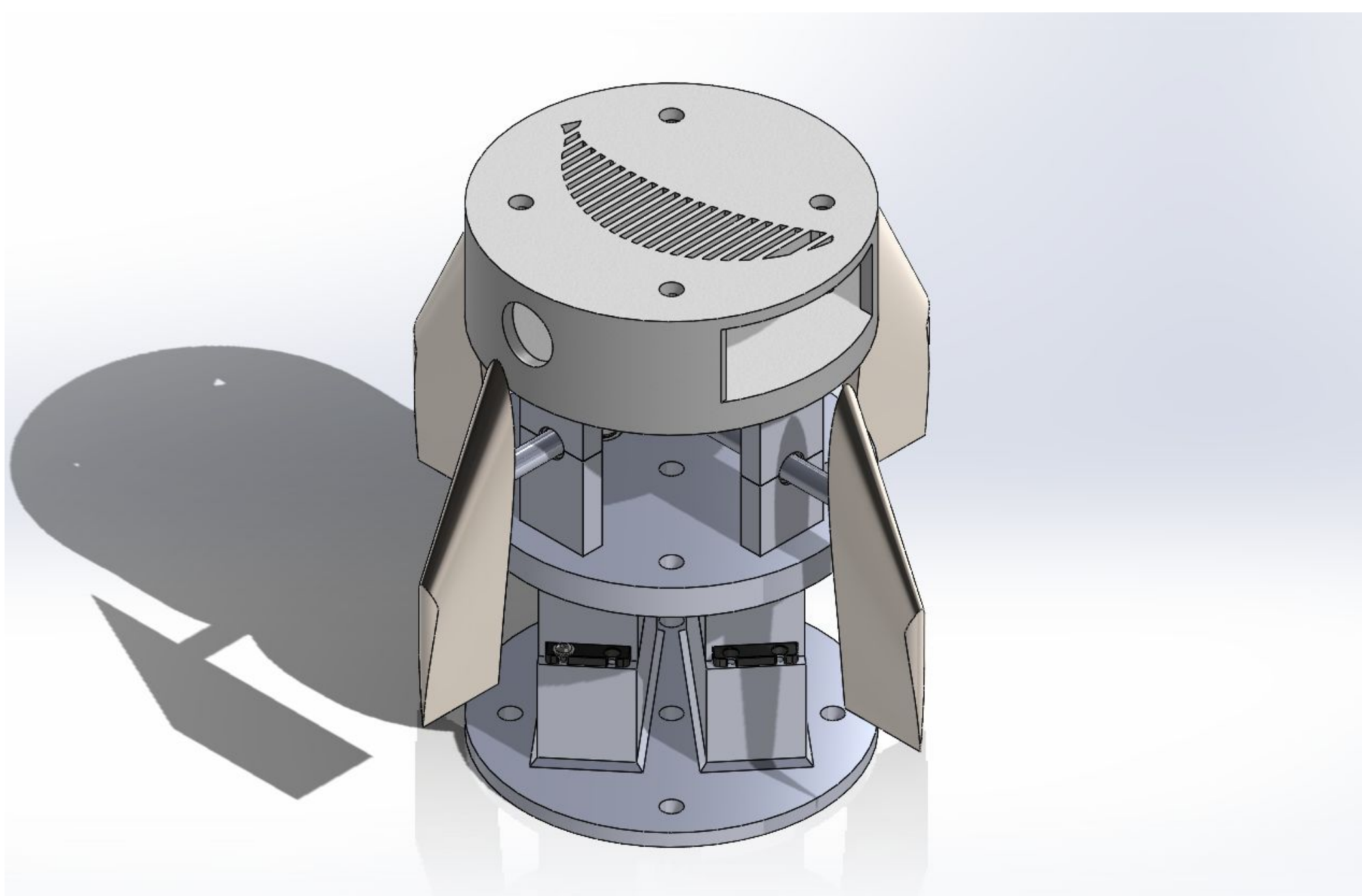
Rocket Navigation Control System

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Abstract

A rocket's Reaction Control System (RCS) is a system that has dynamic fins that are controlled by motors. These motors rotate fins in order to keep the rocket in a straight path and reduce the roll of the rocket and minimize the chance of the rocket overturning.

Background & Introduction



The RCS consists of dynamic fins connected to servo motors through a gear drive train, which are controlled by a microcontroller using a PD controller. The system is designed to ensure a rocket's path is straight and to prevent it from overturning due to pitch and yaw at subsonic and supersonic speeds. The entire system is housed inside a 3D-printed polycarbonate casing that includes a printed circuit board (PCB), sensors, servos, batteries, beveled gear system, and hardware like bearings and switches. The rods extending from the casing through the rocket body tube walls attach to the exterior fins for effective control of the rocket's movement. These fins are made of carbon fiber with a polyurethane foam core, and are fixed to rotate via adapters on $\frac{3}{8}$ " steel shafts.

Methods

Stability of the rocket is based on three values of the rocket; the center of pressure, center of gravity, and diameter of the rocket. The Barrowman Equations are equations which can take these three values into account, and a simulation program called OpenRocket was used to account for the mass, fin size, and rocket's flight. This program will also take into account the rocket motor's properties, the environment, velocity, acceleration, and more. This can further be information that is useful for the PCB, control loop, and more properties that are of significance to the RCS system. Additional simulations in ANSYS and Solidworks for drag and stress on the fins will aid in these considerations. A constant force of 469 N per fin to simulate the drag force that the rocket will be experiencing during flight

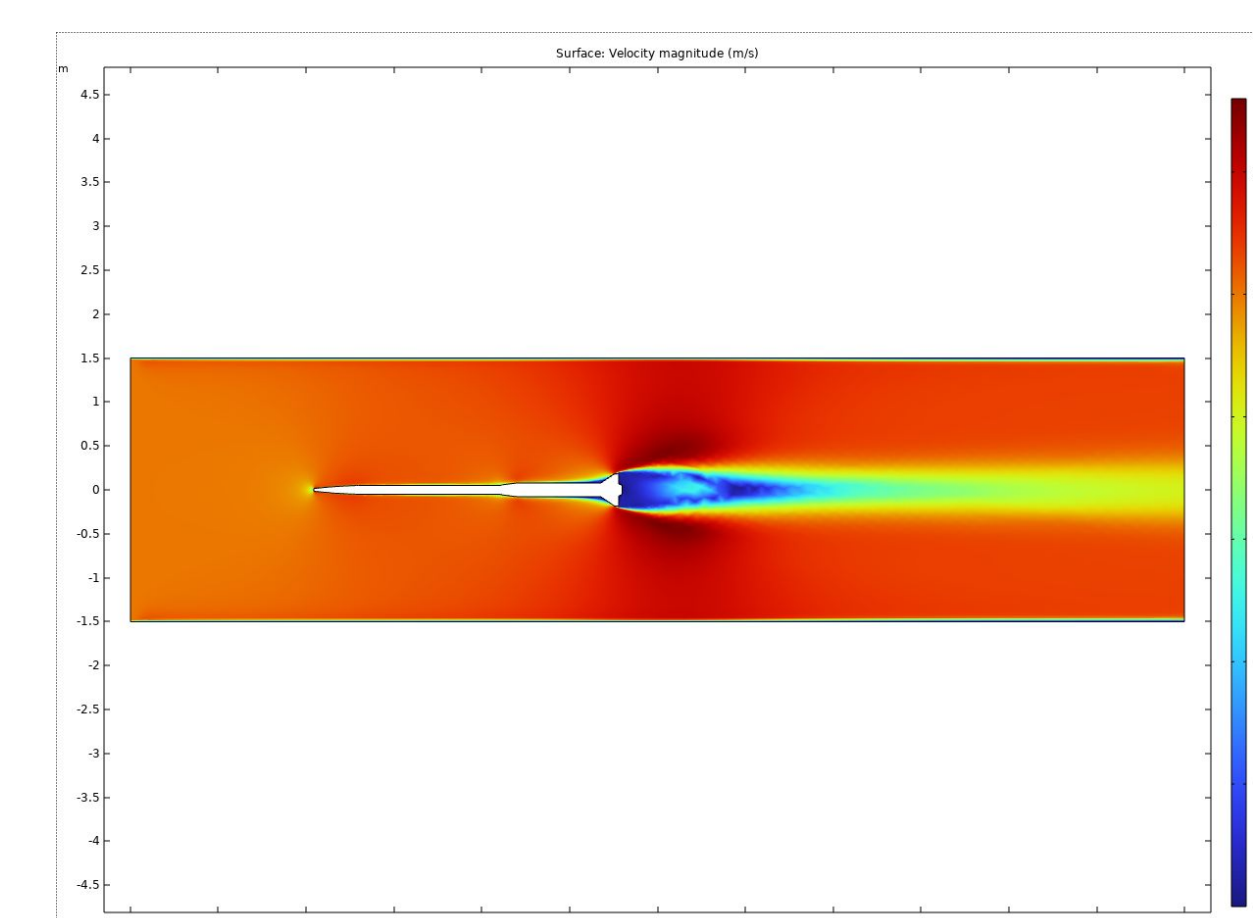
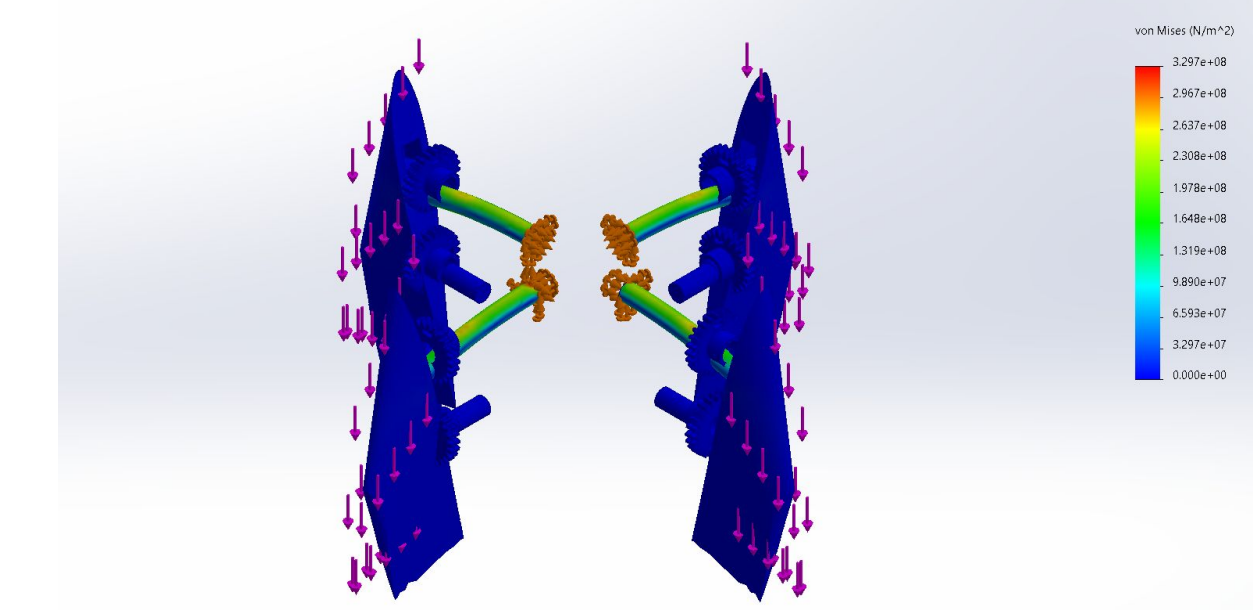
In order to begin assembling the RCS, we decided to follow a modular "stacking" format. This included specific layers in the housing for the servos, rod system, and PCBs. In this sense, the layers can easily be swapped out and added onto. These housings were chosen to be 3D printed, as the printers are easily accessible, and therefore a new layer can be made rapidly if needed. The materials used were 3D printed polycarbonate, as well as carbon fiber fins for the light weight and strength for a high-powered rocket flight.

Results

The PD controller loop controls 2 pair of fins for both the pitch and yaw moments of the rocket. In simulations, the fin design shows an ideal lift:drag ratio over the course of the flight. The polyurethane foam core was successfully cut out and proved to be a sturdy material for the wet layup process. The wet layup process was successful with a proper vacuum seal, as well as an inner and outer mold to hold the shape of the fin itself. Additionally, the composite structure passes all structural tests that it has been subjected to, and was successfully mounted to the RCS system itself.

Simulations were completed for drag force on the fins in Solidworks and ANSYS, however, flow for turbulent models are continuing to be researched and will be handed off to the next group continuing the project. An ANSYS laminar flow simulation for velocity and heat flux was successful, and can be worked off of in the future. With a simulation on Mises stress on the fins, it is observed that the highest stress appears on the rods supports that secure the ball bearings to the housing.

For initial testing, the RCS is mounted upon a cord that is strung up. It is then pulled back and released so that it swings freely. Upon release, the system will actively correct the course of the system so that it flies in a straight path in the direction of the body tube rather than sideways. This setup was chosen to demonstrate the system's ability to use real time data to correct its course while in a smaller environment. As this would ideally be used to launch a rocket, this smaller demonstration is needed. After this, the next step in testing is to implement the RCS in a model rocket launch.



Design

The RCS uses electronic hardware components and servo motors to control the fins of a rocket. The subsystem has two 100 kg holding force motors powered by a 4 cell 2250 mAh Lipo battery, and a collection of boards and sensors powered by a 3 cell 450 mAh variant. The system uses an altimeter/barometric pressure sensor and three orientation sensors to accurately orient the fins, which continuously adjusts the orientation through a PD loop. The onboard microcontroller is connected to a servo suite via a custom-designed four-layer PCB.

A basic supersonic airfoil was chosen as best suited for supersonic flight. A carbon fiber wet layup was performed on the fin with a polyurethane foam core. The primary housing structure has three circular plates made of polycarbonate that fit inside the rocket body tube. The bottom housing section contains two servo motors attached to vertical rods that pass through the center housing section where pinion gears mate with beveled gears for a 2:1 ratio to increase torque. The larger gears drive $\frac{3}{8}$ " horizontal shafts made from 303 stainless steel, attached to carbon fiber fins via adapters. The design allows for scaling of the system by reprinting the polycarbonate disks. Future additions such as GPS and Bluetooth modules and antennas can be accommodated by the custom-designed PCB.

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